

OpenClaw API Key Configuration

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1. Course Overview

This course provides a detailed introduction to configuring AI model API keys for OpenClaw, including the API key application process and two configuration methods (CLI command and configuration file). It also covers how to verify whether the configuration is successful. After completing this course, you will be able to successfully connect to and use various AI large model service providers.

1.1 Course Content

- Understand the types of model service providers supported by OpenClaw
- Master the API key application method
- Learn to view and verify the status of connected models

2. API Key Application

2.1 Recommended Model Service Providers

- **Recommended for beginners:** Refer to the accompanying tutorial [Chapter 11 Online AI Large Model Development] — [2. Tongyi Qianwen Large Model] to apply for the corresponding model provider API key
- OpenClaw supports numerous model service providers, including:
 - **OpenAI** - GPT series (API + Codex)
 - **Anthropic** - Claude series (API + Claude Code)
 - **Google** - Gemini series

- **Alibaba Cloud** - Qwen Cloud, Tongyi Qianwen
- **DeepSeek** - DeepSeek models
- **Zhipu AI** - GLM models
- **Moonshot AI** - Kimi + Kimi Coding
- **Ollama** - Local models (cloud + local)
- And many other service providers
- You can freely choose a model service provider based on your actual needs and the network policies of your region, register on the corresponding official website, and apply for an API key

3. Configuring OpenClaw API Key

3.1 Configuration Methods

OpenClaw provides two ways to configure the API key:

1. **CLI command configuration** (suitable for beginners)
2. **Directly modifying the configuration file** (suitable for developers)
 - Configuration file path: `$HOME/.openclaw/openclaw.json`

3.2 Quick Configuration

- Suitable for users who want a quick start experience. You first need to register an Alibaba Cloud Bailian account and create an API key (refer to the accompanying tutorial [Chapter 11 Online AI Large Model Development] — [2. Tongyi Qianwen Large Model])
- The factory preset configuration file already has Bailian model parameters configured; just fill in the API key to use it directly
- Open the OpenClaw configuration file:

```
nano $HOME/.openclaw/openclaw.json
```

3.2.1 Configure baseUrl

- **Users in mainland China do not need to modify this; skip this step.** For overseas users, select the appropriate baseUrl for your service region (go to the [Alibaba Cloud Bailian Console](#) homepage, switch to the target region in the upper right corner of the page, such as **Singapore**, then select **API Key** in the left navigation bar to enter the management page and create an API Key)

Base URL for the API Key's region:

Singapore: `https://dashscope-intl.aliyuncs.com/compatible-mode/v1`

US (Virginia): `https://dashscope-us.aliyuncs.com/compatible-mode/v1`

3.2.2 Fill in apiKey

```
"models": {  
  "mode": "merge",  
  "providers": {  
    "bailian": {  
      "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",  
      "apiKey": "sk-a87e1137b17b4a35af13aa4a12ac3473",
```

- After filling in, press `Ctrl+X`, then press `Y` to confirm saving, and press Enter to exit

4. View Connected Model Status

4.1 View Configured Model List

After configuration is complete, you can use the following command to view connected models:

```
openclaw models list
```

```
pi@raspberrypi:~$ openclaw models list
🐉 OpenClaw 2026.3.12 (6472949) - Your second brain, except this one actually remembers where you left things.

14:49:14 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
14:49:14 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_user_get_messages, feishu_im_user_get_thread_message
s, feishu_im_user_search_messages
14:49:14 [plugins] feishu_search: Registered feishu_search_doc_wiki
14:49:14 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
14:49:14 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
14:49:14 [plugins] feishu_sheets: Registered feishu_sheet
14:49:14 [plugins] feishu_im: Registered feishu_im_bot_image
14:49:14 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
14:49:14 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
14:49:14 [plugins] feishu_oauth: Registered feishu_oauth tool
14:49:14 [plugins] feishu_oauth_batch_auth: Registered feishu_oauth_batch_auth tool
14:49:14 [plugins] Pi Control extension registered: servo, dht, rgb, oled, delay, camera (direct CLI)
14:49:14 [plugins] pi-control: Feishu config loaded (chatId=oc_55f4adf9c958ff43c1083a138aa5e18e)

Model          Input  Ctx   Local Auth  Tags
bailian/qwen3.5-plus  text+image  977k  no  yes  default,configured
bailian/MiniMax-M2.5  text        192k  no  yes  configured
bailian/glm-4.7       text        198k  no  yes  configured
pi@raspberrypi:~$
```

4.2 Verify Configuration Success

1. Test conversation

```
openclaw tui
```

Launch the TUI interface, send a test message (e.g., "Hello"), and confirm that the model can respond normally.

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)

|
```

5. Other Connection Methods (Advanced)

- If you have your own model provider or want to connect to other providers later, refer to the following tutorial

5.1 CLI Command Configuration API Key (Suitable for Beginners)

1. Launch the configuration wizard

```
openclaw onboard
```

- Select [Yes], press Enter

```
Recommended baseline:
- Pairing/allowlists + mention gating.
- Multi-user/shared inbox: split trust boundaries (separate
  gateway/credentials, ideally separate OS users/hosts).
- Sandbox + least-privilege tools.
- Shared inboxes: isolate DM sessions (`session.dmScope:
  per-channel-peer`) and keep tool access minimal.
- Keep secrets out of the agent's reachable filesystem.
- Use the strongest available model for any bot with tools or
  untrusted inboxes.
```

```
Run regularly:
openclaw security audit --deep
openclaw security audit --fix
```

```
Must read: https://docs.openclaw.ai/gateway/security
```

```
◆ I understand this is personal-by-default and shared/multi-user use requires
lock-down. Continue?
  ● Yes / ○ No
```

- Select QuickStart for configuration mode, press Enter

```
◆ Setup mode
  ● QuickStart (Configure details later via openclaw configure.)
  ○ Manual
```

2. Select model service provider

- Use the up and down arrow keys to select the service provider you got an API key from, and press Enter to confirm. Here we use Alibaba Cloud Bailian model provider as an example; the configuration method for other model providers is the same.
- Here select custom provider **[Custom Provider]**

```
◆ Model/auth provider

Search:
  ○ Alibaba Model Studio
  ○ Anthropic
  ○ Arcee AI
  ○ BytePlus
  ○ Chutes
  ○ Cloudflare AI Gateway
  ○ Copilot
  ● Custom Provider (Any OpenAI or Anthropic compatible endpoint)
```

- Fill in the model provider's base_url

```
◆ API Base URL
  https://dashscope.aliyuncs.com/compatible-mode/v1
```

3. Enter API Key

- After selecting the service provider, the system will prompt you to enter the corresponding API key

```
◆ How do you want to provide this API key?  
● Paste API key now (Stores the key directly in OpenClaw config)  
○ Use external secret provider
```

- Paste or enter the API key you applied for, and press Enter to confirm

```
◆ API Key (leave blank if not required)  
sk-a87e1137b17b4a35af13aa4a1xxxxxx
```

- Select **OpenAI-compatible** for the interface protocol

```
◆ Endpoint compatibility  
● OpenAI-compatible (Uses /chat/completions)  
○ Anthropic-compatible  
○ Unknown (detect automatically)
```

4. Fill in the model name; here we use MiniMax as an example

```
◆ Model ID  
MiniMax-M2.5
```

- **Endpoint ID** is the model provider identifier; you can enter any custom identifier

```
◆ Endpoint ID  
custom-dashscope-aliyuncs-com
```

- Model alias (optional) can be left blank

```
◆ Model alias (optional)  
e.g. local, ollama
```

5. Other configurations

- For other configuration items, select 'Skip' for all items as shown in the figure below; no additional configuration is needed

```
◇ Skills status

Eligible: 16
Missing requirements: 38
Unsupported on this OS: 7
Blocked by allowlist: 0

◇ Configure skills now? (recommended)
○ Yes / ● No
```

- For Enable Hooks, press the spacebar to select **Skip for now**

```
Enable hooks?
■ Skip for now
□ 🚀 boot-md
□ 📎 bootstrap-extra-files
□ 📝 command-logger
□ 💾 session-memory
```

- Select Restart to restart the gateway and apply the configuration

```
Gateway service already installed
● Restart
○ Reinstall
○ Skip
```

5.2 Modify Configuration File (Recommended Method)

Configuration File Location

The configuration file is located at: `$HOME/.openclaw/openclaw.json`

Edit Configuration File

1. Open the configuration file

```
nano $HOME/.openclaw/openclaw.json
# Or use your preferred editor, such as vim, code, etc.
```

2. Add model provider configuration

Add or modify the `models.providers` section in the configuration file:

```
{
  "models": {
    "providers": {
      "openai": {
        "apiKey": "OPENAI_API_KEY"
      },
      "anthropic": {
        "apiKey": "ANTHROPIC_API_KEY"
      },
      "google": {
        "apiKey": "GOOGLE_API_KEY"
      }
    }
  },
  "agents": {
    "defaults": {
      "model": {
        "primary": "openai/gpt-4"
      }
    }
  }
}
```

3. Configuration description

- `models.providers`: Configure the API keys for each model service provider
 - `openai.apiKey`: OpenAI's API key
 - `anthropic.apiKey`: Anthropic's API key
 - `google.apiKey`: Google's API key
- `agents.defaults.model.primary`: Set the default model to use
 - Format is `provider/model-name`, e.g., `openai/gpt-4`, `anthropic/claude-3`, `google/gemini-pro`

4. Save and exit

- Press `Ctrl+O` to save, `Ctrl+X` to exit (nano editor)

5. Factory default Bailian configuration (can be directly modified or referenced)

```
"models": {
  "mode": "merge",
  "providers": {
    "bailian": {
      "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",
      "apiKey": "sk-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
      "api": "openai-completions",
      "models": [
        {
          "id": "qwen3.5-plus",
          "name": "qwen3.5-plus",
          "reasoning": false,
          "input": [
            "text",
            "image"
          ]
        }
      ]
    }
  }
}
```

```

    ],
    "contextwindow": 1000000,
    "maxTokens": 65536,
    "cost": {
      "input": 0,
      "output": 0,
      "cacheRead": 0,
      "cachewrite": 0
    }
  },
  {
    "id": "MiniMax-M2.5",
    "name": "MiniMax-M2.5",
    "reasoning": false,
    "input": [
      "text"
    ],
    "cost": {
      "input": 0,
      "output": 0,
      "cacheRead": 0,
      "cachewrite": 0
    },
    "contextwindow": 196608,
    "maxTokens": 32768
  },
  {
    "id": "glm-4.7",
    "name": "glm-4.7",
    "reasoning": false,
    "input": [
      "text"
    ],
    "cost": {
      "input": 0,
      "output": 0,
      "cacheRead": 0,
      "cachewrite": 0
    },
    "contextwindow": 202752,
    "maxTokens": 16384,
    "compat": {
      "thinkingFormat": "qwen"
    }
  }
]
}

```

6. Common Problems and Solutions

6.1 API Key Invalid or Error

Problem: After configuration, the system reports that the API key is invalid or authentication has failed

Solution:

1. Confirm whether the API key is correct (make sure there are no extra spaces)
2. Check whether the API key has expired or been disabled
3. Check your account balance

6.2 Unable to Access Model Service

Problem: Configuration is correct but cannot connect to the model service

Solution:

1. Check whether the network connection is normal
2. Confirm whether the network in your region can access the service provider (some services may require special network configuration)
3. Try switching to another model service provider for testing

7. Course Summary

Through this course, you have mastered:

- ☒ Types of model service providers supported by OpenClaw
- ☒ API key application methods and precautions
- ☒ CLI command configuration method (suitable for beginners)
- ☒ Configuration file modification method (suitable for developers)
- ☒ Methods for viewing and verifying model configuration status
- ☒ Troubleshooting and solutions for common problems